Oz Amram

AI Safety Researcher • Fellow at CBAI • Background in particle physics

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Research interests: personas, subliminal learning, mechanistic interpretability.

AI Safety Research

AI Safety Research Fellow, Cambridge Boston Alignment Initiative (CBAI) Oct 2026–present

- Mentor: Hadas Orgad. Testing whether models can evade activation-based probes, with the goal of building a benchmark for this capability.

O. **Amram**. "Base Models Can Detect Hidden Intentions that Prompted Classifiers Miss" (2026). arXiv: 2610.36202

Sole author; independent work, before CBAI. Introduced a method to audit hidden intentions / subliminal learning signals using "persona-header likelihood evaluation" (PHLE): score a model's responses under a base model conditioned on a one-paragraph description of a hypothesized hidden intention. Detects a concealed intent to harm the user with high accuracy, where prompted classifiers remain near chance. Reads subliminal signals in number sequences, transfers across model families and survives distillation. Earlier version on LessWrong. Code: github.com/OzAmram/psm-mixture-test.

Physics Research Positions

Wilson Fellow & Associate Scientist (tenure-track), Fermilab (on leave since Oct 2026) 2026

Postdoctoral Researcher, Fermilab 2022–2026

- Research on **machine learning for particle physics**; member of the CMS experiment at the Large Hadron Collider (CERN).
- Co-convener, CMS exotic-searches-with-jets analysis group (EXO Jets+X), 2024–6: review and manage $\sim$15 active analyses (papers), 15 public results during tenure.
- **Anomaly detection**: Led first CMS paper using ML-based anomaly detection to search for new particles: five complementary unsupervised methods applied to 30M collision events, with up to sixfold gains in discovery sensitivity over traditional methods across a wide range of signal types never seen in training. Designed the search's statistical and interpretability frameworks to characterize what makes a flagged event anomalous and map it back to physical detector signatures.
- **Generative models**: Wrote CaloDiffusion: up to $1000\times$ faster than classic physics simulations, near indistinguishable quality. Led the first application of generative models to simulation in CMS: a generative model with 500k-dimensional output for the High-Granularity Calorimeter.
- **Mentored** students at UChicago, UCSD, UCSB, Cornell, KIT, JHU and Hamburg on anomaly-detection and generative modeling projects.

Education

Ph.D., M.A. Physics, Johns Hopkins University 2016–2022

- Thesis: *Searching for Anomalies in Proton-Proton Collisions at the Large Hadron Collider*. Advisors: Morris Swartz, Petar Maksimovic.
- Independently developed Tag N' Train, an anomaly-detection algorithm and a top performer in the LHC Olympics community competition.

B.S. Physics, minor Philosophy, Carnegie Mellon University 2012–2016

Selected Physics Publications

18 physics publications (11 as primary author or lead contributor); 701 citations, h-index 9 (423 and 8 excluding self-citations). Additionally an author on 400+ CMS collaboration papers.

1. O. **Amram**, M. Letizia, and M. Kuusela. "Model-Agnostic Signal Discovery with Machine Learning: Bridging the Gap Between Theory and Practice" (2026). arXiv: 2605.31103

   Lead author. Review paper on anomaly detection, covering the statistical basis, applications and validation strategies. Invited contribution to a series on AI in particle physics.
2. CMS Collaboration. "Model-agnostic search for dijet resonances with anomalous jet substructure in proton-proton collisions at $\sqrt{s}$ = 13 TeV". *Rept. Prog. Phys.* (2025). arXiv: 2412.03747

   Lead author (see Research Positions). Selected as an IOP Research Highlight.
3. O. **Amram** and M. Szewc. "Data-Driven High-Dimensional Statistical Inference with Generative Models". *JHEP* (2025). arXiv: 2506.06438

   Lead author. Replaces binned histogram fits with a learned generative model, enabling likelihood fits in 5+ dimensions; $\sim$40% improvement on a benchmark measurement. Code: github.com/OzAmram/HI-SIGMA.
4. O. **Amram** et al. "Aspen Open Jets: unlocking LHC data for foundation models in particle physics". *Mach. Learn. Sci. Tech.* (2025). arXiv: 2412.10504

   Built and released the dataset and its processing tooling: 180M jets (200 GB) derived from public collider data, to support pretraining of foundation models on real rather than simulated data. Code: github.com/OzAmram/AOJProcessing.
5. O. **Amram** and K. Pedro. "Denoising diffusion models with geometry adaptation for high fidelity calorimeter simulation". *Phys. Rev. D* (2023). arXiv: 2308.03876

   Lead author; wrote the architecture, noise schedule and sampler from scratch. Geometry-adapted denoising diffusion over high-dimensional structured data; achieved best-in-class sample fidelity under the physics-based evaluation metrics of the community CaloChallenge benchmark. Code: github.com/OzAmram/CaloDiffusion.
6. O. **Amram** and C. M. Suarez. "Tag N' Train: a technique to train improved classifiers on unlabeled data". *JHEP* (2021). arXiv: 2002.12376

   Independently developed. Co-training scheme for particle physics data that learns from noisy, mixed-sample supervision with no labeled examples of the target class. Code: github.com/OzAmram/TagNTrain.

Technical Skills

- **Languages:** Python (primary, expert). C++.
- **Frameworks:** PyTorch (primary); TensorFlow/Keras.
- **Language models:** likelihood scoring with open-weight base and instruct models; LoRA fine-tuning; prompted-classifier baselines via API. Training activation-based probes / deriving concept vectors.
- **Methods:** diffusion models; weakly supervised, co-training, autoencoders and unsupervised methods for anomaly detection; generative models for fast simulations and likelihood-based inference; uncertainty quantification.

Awards & Funding

- **DOE Genesis Mission grant** (2026): *AIDA-Scout*, real-time anomaly detection on the full LHC collision rate. Originated the idea and wrote the majority of the successful proposal; senior/key personnel. $135k Fermilab share of $660k across five institutions, nine months from August 2026.
- Fermilab LHC Physics Center Distinguished Researcher (2026).
- Rising Star in Experimental Particle Physics, University of Chicago (2021).

Additional

- **Talks:** 21 conference talks (7 invited); 11 invited seminars incl. MIT, SLAC, UChicago.
- **Teaching:** Founding organizer of the annual Machine Learning for Fundamental Physics school ($\sim$40 in person, $\sim$100 virtual); gave the anomaly-detection lecture and ran the tutorial. Regular teacher at annual CMS data analysis school at Fermilab.
- **Writing:** $\sim$20 Particle Bites articles for a general audience; personal blog; CMS and CERN news articles on the anomaly search.
- **Reviewing:** NeurIPS Machine Learning for Physical Sciences workshop (2025–6); ACAT (2023). Journal referee for SciPost Physics, Physical Review D, Physics Letters B, Scientific Reports.